

This by no means implies or excludes similar questions or marks distribution on the exam.

1. (8 marks) Use the rules of inference to show that the following argument is valid. Provide the rule for each step.

	STEPS	REASON
p	1) q	given
$q \rightarrow r$	2) $q \rightarrow r$	given
$(p \wedge q) \rightarrow (s \vee \neg r)$	3) r	M.P. 1, 2
q	4) p	given
$\therefore \neg s \rightarrow u$	5) $p \wedge q$	R.C. 1, 4
	6) $(p \wedge q) \rightarrow (s \vee \neg r)$	given
	7) $s \vee \neg r$	M.P. 5, 6
	8) s	D.S. 3, 7
	9) $s \vee u$	D.A. 8
	10) $\neg s \rightarrow u$	No NAME 9
	11)	
	12)	
	13)	
	14)	

2. (6 marks) Provide one example of nonempty sets A, B and C that satisfies all of the following:

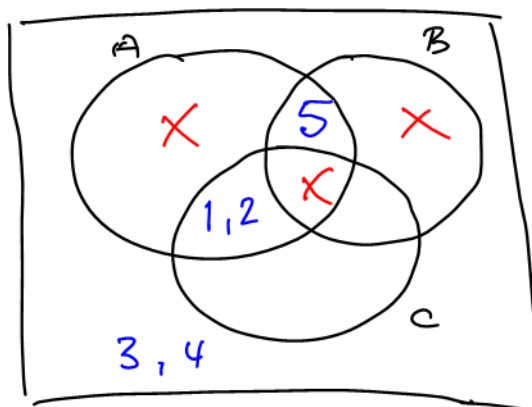
- $|A \cap (\overline{B \cup C})| = 0$

- $(A \cap B) \subseteq (B - C)$

- $(B - C) \subseteq A$

- $5 \in (A \cap B)$

- $(A \Delta B) \cap C = \{1, 2\}$



Universe $U = \{1, 2, 3, 4, 5\}$

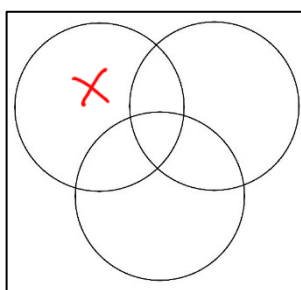
Write your answer:

$A = \{ \quad 1, 2, 5 \quad \}$

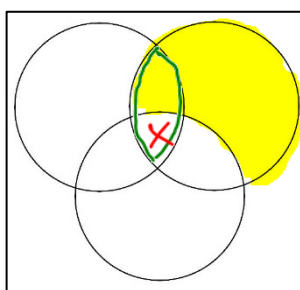
$B = \{ \quad 5 \quad \}$

$C = \{ \quad 1, 2 \quad \}$

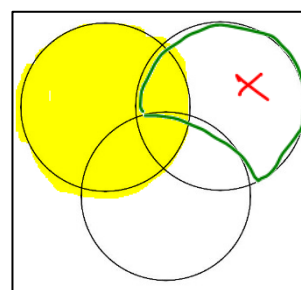
Must write your answer here!
Leaving at least one empty
implies zero marks.



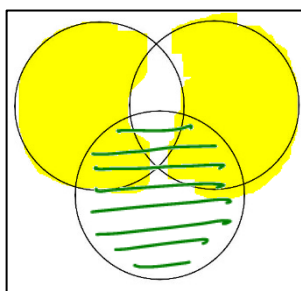
$|A \cap \overline{B \cup C}| = 0$



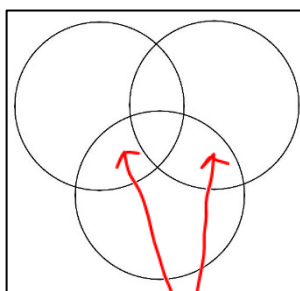
$A \cap B \subseteq B - C$



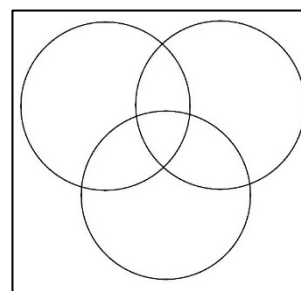
$(B - C) \subseteq A$



$\{1, 2\} = (A \Delta B) \cap C$



only 1, 2
are here



3. [16 marks] The System Administrator has set the following rules for the password:

- ✓ The password is a string made of 8 characters.
- ✓ The available characters are 26 upper case letters of English alphabet A, B, C... Z and 10 digits 0, 1... 9.

a) [2 marks] How many passwords have letters only?

$$\underline{\underline{26^8}}$$

b) [2 marks] How many passwords with letters only have no repeated letters?

$$P(26, 8) = \frac{26!}{18!}$$

c) [3 marks] How many passwords have five letters "C", two letters "B" and a digit?

$$\frac{8!}{5!2!} \left\{ \begin{array}{l} C C C C C B D \quad (10) \\ C C C \quad (10) B C B C \end{array} \right. \quad \therefore \frac{8!}{5!2!} \cdot (10)$$

d) [3 marks] The password is considered weak if it contains letters only or digits only. How many passwords have at least one digit and at least one letter?

$$36^8 - 26^8 - 10^8$$

e) [3 marks] How many passwords with digits only have non-decreasing sequence of digits and start with 5? The sequence of eight digits $D_1 D_2 D_3 D_4 D_5 D_6 D_7 D_8$ is non-decreasing if $D_1 \leq D_2 \leq D_3 \leq D_4 \leq D_5 \leq D_6 \leq D_7 \leq D_8$.

sort - think combinations

size $r=7$
digits 5, 6, 7, 8, 9 so $n=5$

$$D_1 \leq D_2 \leq \dots \leq D_8 \quad \text{with repetition allowed} \quad \binom{n+r-1}{r} = \binom{5+7-1}{7} = \binom{11}{7}$$

f) [3 marks] How many passwords have three letters "A", two letters "B", two letters "C" and one letter "D" such that no A's are adjacent.

$$\frac{5!}{2!2!} \left\{ \begin{array}{l} \wedge B \wedge B \wedge C \wedge C \wedge D \wedge \quad (6) \\ \wedge B \wedge D \wedge C \wedge B \wedge C \wedge \quad (6) \end{array} \right. \quad \therefore \frac{5!}{2!2!} \binom{6}{3}$$

4. [8 marks] Circle one answer only; you MUST justify the answer. No marks will be given for answers without justification. No marks will be given when justification does not imply the correctness of the answer.

Sets A and B are defined as follows: $A = \{0, 1, 2, 3\}$ and $B = \{-3, -2, -1\}$

a) $\exists x \in A, \exists y \in A$ such that $x + y = 5$

TRUE/FALSE

x	y	$x + y = 5$
2	3	$5 = 5$ ✓

b) $\forall x \in A, \forall y \in B$, then $x \cdot y < 0$

TRUE/FALSE

x	y	$x \cdot y < 0$
0	-3	$0 < 0$ ✗

c) $\exists x \in A, \forall y \in B$ such that $x \cdot y \in A$

TRUE/FALSE

x	y	$x \cdot y \in A$
0	-3	$0 \in A$ ✓
0	-2	$0 \in A$ ✓
0	-1	$0 \in A$ ✓

d) $\forall x \in A, \exists y \in B$ such that $\{x - y\} \subseteq \{-29, 0, 1, x + y\}$

TRUE/FALSE

x	y	$\{x - y\} \subseteq \{-29, 0, 1, x + y\}$
1	-1	$\{2\} \subseteq \{-29, 0, 1, 0\}$ ✗
1	-2	$\{3\} \subseteq \{-29, 0, 1, -1\}$ ✗
1	-3	$\{4\} \subseteq \{-29, 0, 1, -2\}$ ✗

$$\text{NO NAME: } a \rightarrow b \Leftrightarrow \neg a \vee b$$

5a. (2 marks) Negate and simplify the following statement:

De Morgan's

$$\neg(\forall x [p(x) \rightarrow (\neg r(x))]) = \neg(\forall x \neg p(x) \vee \neg r(x))$$

$$= \exists x p(x) \wedge r(x)$$

5b. (2 marks) Write the negation of the following statement:

If $\overset{P}{c \text{ divides } a} \wedge c \text{ does not divide } b$ then $\overset{Q}{c \text{ does not divide } (a+b)}$

$$\neg(P \rightarrow Q) = \neg(\neg P \vee Q) = P \wedge \neg Q, \text{ in English: } c \text{ divides } a \wedge c \text{ does not divide } b \text{ and } c \text{ divides } (a+b)$$

6. (7 marks) Consider the following programming segment:

***You should note that the two for loops are nested loops.

counter = 150

for i = 3 to n do {

counter = counter + 5

for j = i+1 to 5n+3 do {

counter = counter + 10

}

}

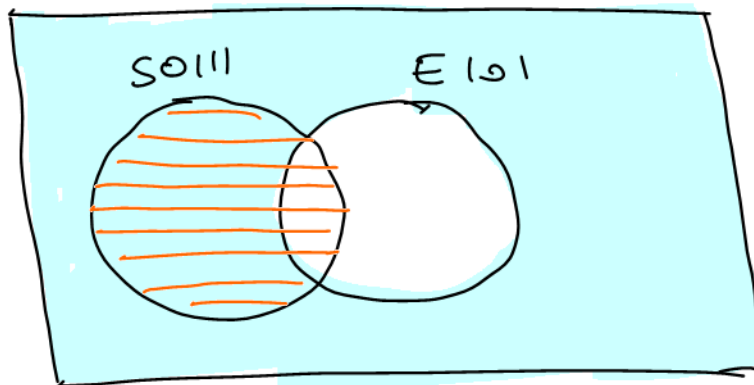
$$\text{note: } (5n+3) - 3 + 1 = n+1$$

$$\text{GAP: } (5n+3) - (n+1) + 1 = 4n+3$$

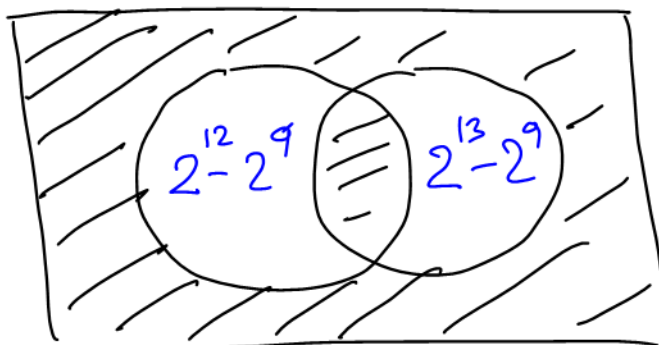
Determine the value that the variable *counter* will have after the segment is executed. Provide your answer in terms of *n*. ***In order to receive marks your approach must be combinatorial - which means that you have to consider combinatorial structures like combinations, permutations or arrangements. Answers that rely on sigma notation and/or solving summation will not be accepted.***

$$C = 150 + (n-3+1)(5) + \left[\binom{n+1}{2} - \binom{4n+3}{2} \right] (10)$$

7. (4 marks) What is the probability that a randomly selected binary string of length 16 starts with 0111 or does NOT end with 101 exclusively ("exclusive or" = one or the other but not the both)?



So, RED or Blue
but not
both



$$Pr = \frac{2^{16} - (2^{12} - 2^9) - (2^{13} - 2^9)}{2^{16}}$$